

Product Requirements Document (PRD) for SENA Network

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1. Introduction

1.1 Purpose

This Product Requirements Document (PRD) outlines the features, functionalities, and user experience for the SENA Network, a Layer 2 (L2) modular infrastructure. It serves as a guiding document for product managers, designers, engineers, and other stakeholders, ensuring a shared understanding of what needs to be built and why. This PRD complements the previously created Project Proposal, Software Requirements Specification (SRS), and Software Design Description (SDD) by focusing on the product from a market and user perspective.

1.2 Scope

The SENA Network aims to deliver a high-performance, secure, and user-friendly L2 blockchain solution that abstracts away Web3 complexities for consumer applications. This PRD covers the product vision, target users, key features, and success metrics for the initial launch and subsequent iterations. It specifically addresses the core L2 protocol, its key modules (Keyless Authentication, Social Connect, Multi-Asset Gas Paymaster, Optimistic Governance), and the integration of the “Aptos-Level Secure” model.

1.3 Definitions, Acronyms, and Abbreviations

Refer to Section 1.3 of the Software Requirements Specification (SRS) for a comprehensive list of definitions, acronyms, and abbreviations.

1.4 References

- [1] SENA Network: Layer 2 Modular Infrastructure Documentation (Stakeholder Document)
- [2] Software Requirements Specification (SRS) for SENA Network
- [3] Software Design Description (SDD) for SENA Network
- [4] Aptos Keyless Accounts. *Aptos Docs*. Available at:
<https://aptos.dev/concepts/accounts/#keyless-accounts>

2. Product Vision

SENA Network envisions a future where blockchain technology is seamlessly integrated into everyday consumer applications, making decentralized services as accessible and intuitive as their Web2 counterparts. We aim to be the foundational L2 infrastructure that empowers developers to build next-generation dApps without compromising on user experience, security, or scalability.

3. Target Users and Use Cases

3.1 Target Users

- **Web2 Developers:** Developers familiar with traditional web development paradigms (e.g., JavaScript, Python) who want to build on blockchain without deep Web3 expertise.
- **Consumer Application Developers:** Teams building social media, gaming, fintech, or e-commerce applications that require high transaction throughput, low latency, and a frictionless user experience.
- **End-Users:** Everyday consumers who expect simple onboarding, familiar authentication methods (e.g., Google, Apple login), and predictable transaction costs.

3.2 Use Cases

- **Social dApps:** Users can log in with their existing social accounts, connect with friends using human-readable identifiers, and engage in on-chain activities without managing private keys or gas tokens.
- **Gaming:** Players can interact with in-game assets and economies with instant transactions and predictable fees, using their preferred stablecoins.
- **Fintech & Payments:** Applications can offer seamless, low-cost global payments and financial services, leveraging stablecoins and familiar authentication methods.
- **Decentralized Identity:** Users can manage their digital identities and data with enhanced privacy and control, linked to their Web2 credentials via ZKPs.

4. Key Features and Functionality

SENA Network will provide the following core product features:

4.1 Frictionless Onboarding & Keyless Accounts

- **Feature:** Users can create and access self-custodial blockchain accounts using their existing Web2 identities (e.g., Google, Apple ID) via OpenID Connect (OIDC).
- **Benefit:** Eliminates the need for seed phrases and complex private key management, drastically reducing onboarding friction and improving user adoption.
- **Technical Implementation:** Leverages Aptos Keyless accounts and Zero-Knowledge Proofs (ZKPs) for secure, privacy-preserving authentication, with ZKP generation offloaded to a decentralized prover network.

4.2 Human-Readable Social Identifiers

- **Feature:** An embedded system that maps human-readable identifiers (e.g., usernames, email addresses) to SENA network addresses.
- **Benefit:** Enables intuitive social connections and payment experiences, similar to Web2 platforms, without exposing sensitive personal data.
- **Technical Implementation:** Utilizes privacy-preserving cryptographic hashing (SHA-256 with network salt) for storing identifiers, with RPC endpoints for address resolution.

4.3 Multi-Asset Gas Abstraction

- **Feature:** Users can pay for transaction fees using a whitelist of stablecoins (e.g., USDC, USDT) instead of a native volatile token.
- **Benefit:** Provides predictable and stable transaction costs, removing a major barrier for mainstream users and simplifying application development.
- **Technical Implementation:** The L2 protocol includes a Gas Paymaster Engine that dynamically evaluates gas costs, converts them to stablecoin equivalents, and manages an L1 bridge for continuous funding of L1 commitments.

4.4 Robust Security via Aptos L1 Inheritance

- **Feature:** SENA L2 inherits the security and finality guarantees of the Aptos L1 blockchain.
- **Benefit:** Ensures that user funds are safe and recoverable even in the event of an L2 server compromise, providing an “Aptos-Level Secure” environment.
- **Technical Implementation:** Periodic submission of L2 State Roots to an Aptos L1 Move contract, enabling emergency withdrawals directly from L1.

4.5 Efficient and Transparent Governance

- **Feature:** A low-overhead governance mechanism allowing \$SENA token holders to propose and vote on network parameter changes.
- **Benefit:** Ensures decentralized control and adaptability of the network, with transparent and auditable decision-making.
- **Technical Implementation:** Off-chain aggregation of votes with on-chain validation via a Governance Multi-Signature Council, applying updates to system variables within the Sovereign SDK runtime.

5. User Stories

5.1 End-User Stories

- **As a new user**, I want to sign up for a dApp using my Google account so I don't have to manage a seed phrase.
- **As a user**, I want to send money to my friend using their social handle so I don't have to copy a long address.
- **As a gamer**, I want to buy an in-game item using USDC so I don't have to worry about volatile gas fees.
- **As a user**, I want to be confident that my funds are safe even if the L2 network experiences issues, knowing I can withdraw them from Aptos L1.

5.2 Developer Stories

- **As a dApp developer**, I want to integrate keyless authentication easily so I can onboard Web2 users without friction.
- **As a dApp developer**, I want to use human-readable identifiers for payments and social features so I can build intuitive user experiences.
- **As a dApp developer**, I want my users to pay gas fees in stablecoins so they don't need to acquire a native token.
- **As a dApp developer**, I want to build on a secure L2 that leverages the robust security of Aptos L1.

6. Success Metrics

- **User Adoption:** Number of unique active users and applications deployed on SENA.
- **Transaction Volume:** Total number and value of transactions processed on the L2.
- **Developer Engagement:** Number of developers building on SENA, SDK downloads, and community activity.
- **Security Audits:** Successful completion of independent security audits with minimal critical findings.
- **Performance:** Consistent achievement of target transaction throughput and finality metrics.
- **L1 Commitments:** Regular and successful submission of State Roots to Aptos L1.

7. Future Considerations

- **Decentralized Sequencer:** Explore options for decentralizing the sequencer to enhance censorship resistance and fault tolerance.
- **Fraud Proofs:** Implement a more advanced fraud proof system for optimistic rollup security.
- **Cross-Chain Interoperability:** Investigate integrations with other L1s and L2s to expand ecosystem reach.
- **Developer Tooling:** Enhance SDKs, documentation, and developer support to foster a thriving ecosystem.

8. Conclusion

SENA Network is positioned to revolutionize the adoption of blockchain technology by providing a user-centric and developer-friendly L2 infrastructure. By prioritizing seamless user experience, robust security, and efficient governance, SENA will enable a new generation of decentralized applications that can compete with and surpass their Web2 counterparts.

References

[1] SENA Network: Layer 2 Modular Infrastructure Documentation. *Manus AI*. [2] Software Requirements Specification (SRS) for SENA Network. *Manus AI*. [3] Software Design Description (SDD) for SENA Network. *Manus AI*. [4] Aptos Keyless Accounts. *Aptos Docs*. Available at: <https://aptos.dev/concepts/accounts/#keyless-accounts> [5] Sovereign SDK. *Sovereign Labs*. Available at: <https://www.sovereign.xyz/> [6] RocksDB. *GitHub*. Available at: <https://github.com/facebook/rocksdb> [7] Zero-Knowledge Proofs. *Wikipedia*. Available at:

https://en.wikipedia.org/wiki/Zero-knowledge_proof [8] OIDC and ZKP in Aptos. *Aptos Labs Blog*. Available at: <https://aptoslabs.com/blog/aptos-keyless-accounts-oidc-zkp>